

ENERGY: RMSE at each forecast step				
Lookback = 168 steps, Forecast horizon = 72 steps				
Forecast step	CNN	LSTM	dCeNN+ELM	dCeNN+ELM+ASP
1.000	289.049	572.489	623.944	619.107
2.000	305.769	447.527	611.457	606.752
3.000	331.740	403.728	620.738	616.285
4.000	355.286	403.451	637.551	633.468
5.000	378.953	419.583	639.735	635.567
6.000	395.814	437.764	640.103	635.491
7.000	411.197	452.637	646.057	641.345
8.000	425.772	463.990	646.634	642.360
9.000	437.679	473.404	641.967	637.916
10.000	446.221	481.775	640.837	636.677
11.000	458.554	489.563	635.927	631.280
12.000	466.592	496.904	629.195	623.742
13.000	474.073	503.550	623.448	617.716
14.000	482.014	509.645	623.569	617.884
15.000	488.096	515.370	637.693	631.510
16.000	491.374	520.874	658.589	651.801
17.000	496.532	525.985	664.195	657.389
18.000	496.424	530.477	665.148	658.512
19.000	502.290	534.211	660.767	654.282
20.000	505.123	537.425	647.449	641.577
21.000	507.263	540.319	632.962	627.679
22.000	507.956	543.211	628.427	623.385
23.000	509.542	546.543	629.960	624.879
24.000	514.533	550.203	637.279	632.480
25.000	512.825	553.807	643.883	639.752
26.000	520.412	556.211	655.472	651.971
27.000	520.412	557.296	671.975	668.578
28.000	525.497	557.320	676.156	672.373
29.000	527.699	556.664	668.247	664.521
30.000	525.320	555.612	658.852	655.589
31.000	531.056	554.319	654.440	651.270
32.000	529.615	552.792	657.408	653.954
33.000	527.555	551.073	678.659	674.997
34.000	530.660	549.573	704.286	700.677
35.000	528.222	548.518	707.247	703.692
36.000	527.821	547.836	693.766	689.804
37.000	525.494	547.335	684.022	679.709
38.000	526.608	546.937	689.942	685.807
39.000	527.520	546.723	707.145	703.099
40.000	534.773	546.976	715.109	710.809
41.000	528.515	547.725	704.222	699.461
42.000	532.317	548.799	696.147	691.273
43.000	536.533	549.895	684.074	679.193
44.000	535.148	550.925	666.563	662.032
45.000	538.637	551.966	654.925	651.100
46.000	536.810	553.210	655.190	652.216
47.000	534.249	555.128	655.152	652.778
48.000	528.743	557.568	653.030	650.988
49.000	528.088	559.990	647.839	645.886
50.000	529.922	561.261	651.957	650.035
51.000	532.701	561.439	668.736	666.697
52.000	531.849	561.439	681.727	679.401
53.000	535.636	562.203	675.651	673.153
54.000	535.786	563.413	657.909	655.430
55.000	537.027	564.370	643.890	641.702
56.000	537.003	564.887	641.547	639.735
57.000	532.068	565.340	651.309	649.492
58.000	535.470	566.258	661.250	659.081
59.000	534.181	567.697	655.670	653.020
60.000	534.787	569.621	643.538	640.529
61.000	536.628	571.848	641.576	638.494
62.000	538.048	574.054	652.222	649.129
63.000	535.723	576.055	667.341	664.004
64.000	534.831	577.943	670.514	666.706
65.000	535.961	579.743	660.581	656.278
66.000	536.922	581.505	657.072	652.514
67.000	534.836	583.244	659.063	654.436
68.000	535.585	585.356	657.851	653.255
69.000	540.078	588.182	656.549	652.132
70.000	538.482	591.780	666.669	662.877
71.000	542.462	596.464	676.599	673.222
72.000	548.362	602.120	677.885	674.575

• This table shows how prediction error changes from the first forecast step to the last forecast step.
• It is especially useful for explaining whether a model degrades smoothly or sharply as it predicts further into the future.
• Lower is better. Large mistakes are penalized more strongly than in MAE.